Program 7-Deploy a Web Application to Kubernetes using Minikube

Directory structure:
— Program7
— nginx-deployment.yaml

Step 1: Start Minikube

minikube start
minikube status

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🔄  Restarting existing docker container for "minikube" ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🌟  Enabled addons: metrics-server, storage-provisioner, dashboard, default-storageclass

❗  /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
    ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$
```

Step 2:Create deployment file

nano nginx-deployment.yaml

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05: ~
 GNU nano 7.2                                         nginx-deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 3:Apply deployment

kubectl apply -f nginx-deployment.yaml
kubectl get deployments
kubectl get pods

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   2/2     2            2           9s
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS        AGE
demo-pod                            1/1     Running   1 (6h26m ago)   7h9m
labeled-pod                         1/1     Running   1 (6h26m ago)   6h46m
nginx-deployment-6f9664446b-sckkm   1/1     Running   0               13s
nginx-deployment-6f9664446b-tj72l   1/1     Running   0               13s
simple-pod                          1/1     Running   1 (6h26m ago)   6h53m
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$
```

Step 4:Expose deployment

kubectl expose deployment nginx-deployment --type=NodePort --port=80

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:~$ kubectl get services
minikube service nginx-deployment
NAME               TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.96.0.1     <none>        443/TCP        19d
nginx-deployment   NodePort    10.97.77.2    <none>        80:30823/TCP   10s
|-----------|------------------|-------------|---------------------------|
| NAMESPACE |       NAME       | TARGET PORT |            URL            |
|-----------|------------------|-------------|---------------------------|
| default   | nginx-deployment | 80          | http://192.168.49.2:30823 |
|-----------|------------------|-------------|---------------------------|
🎉  Opening service default/nginx-deployment in default browser...
```

Step 5:Access app

kubectl get services
minikube service nginx-deployment